Hough Transforms in Computer Vision

Hough Transforms
for
Parametric Shape Detection

The Hough Transform is a robust feature extraction technique used to detect parametric shapes (lines, circles, ellipses, etc.) in images by voting in a parameter space. It is tolerant to noise, occlusion, and partial shapes.

1 Standard Hough Transform (Line Detection)

1.1 Concept

The **Standard Hough Transform (SHT)** detects straight lines in binary edge images. A line in the image space (x, y) is represented in the parameter space using the normal form:

$$\rho = x \cos\theta + y \sin\theta$$

where:

- ρ: perpendicular distance from origin to the line,
- θ: angle of the normal with the x-axis ($0 \leq \theta < \pi$).

Each edge pixel (x_i, y_i) votes for all possible (ρ, θ) pairs satisfying the equation.

1.2 Algorithm

1. Apply edge detection (e.g., Canny) to get binary edge map.
2. Create an accumulator array $A(\rho, \theta)$ (discretized parameter space).
3. For each edge pixel (x_i, y_i):
$$\rho = x_i \cos\theta + y_i \sin\theta \quad \forall\theta \in [0, \pi)$$
Increment $A(\rho, \theta)$.
4. Local maxima in A correspond to detected lines.

1.3 Example

Consider edge points: $(1, 1), (2, 2), (3, 3)$ (on line $y = x$).

$$\rho = x \cos\theta + y \sin\theta = x(\cos\theta + \sin\theta)$$

For $\theta = 45^\circ$, $\cos 45^\circ = \sin 45^\circ = \frac{\sqrt{2}}{2}$:

$$\rho = x \cdot \sqrt{2} \quad \Rightarrow \quad \text{Peak at } (\rho = \sqrt{2}, \theta = 45^\circ)$$

Accumulator Peak: Strong vote at correct parameters.

1.4 Advantages

- Robust to noise and gaps.
- Handles multiple lines.

1.5 Disadvantages

- High memory and computation (2D accumulator).
- Sensitive to discretization of ρ, θ.

2 Probabilistic Hough Transform (PHT)

2.1 Concept

The **Probabilistic Hough Transform** reduces computation by randomly sampling edge points instead of using all pixels.

2.2 Algorithm

1. Randomly select a subset of edge points.
2. Apply SHT only on selected points.
3. Repeat and merge results.

2.3 Example

From 1000 edge points, sample 100 $\rightarrow$ $10\times$ speedup.

2.4 Advantages

- Faster than SHT.
- Suitable for real-time applications.

2.5 Disadvantages

- May miss weak lines.
- Accuracy depends on sampling.

3 Generalized Hough Transform (GHT)

3.1 Concept

The **Generalized Hough Transform** detects arbitrary shapes using a template (R-table).

3.2 R-Table Construction

For a template shape with reference point (x_r, y_r):

1. Choose boundary points.
2. For each point p_i, compute vector to reference: $\vec{r}_i = (x_i - x_r, y_i - y_r)$.
3. Store $(\phi, \alpha, \vec{r})$ where ϕ is gradient direction, α is index.

3.3 Detection

For each edge pixel with gradient ϕ:

$$(x_r, y_r) = (x_i, y_i) - \vec{r}(\phi)$$

Vote in 2D accumulator for (x_r, y_r).

3.4 Advantages

- Detects any known shape.
- Scale/rotation invariant with modifications.

4 Circular Hough Transform (CHT)

4.1 Concept

Detects circles using 3D parameter space:

$$(x - a)^2 + (y - b)^2 = r^2$$

Accumulator: $A(a, b, r)$

4.2 Algorithm

1. Edge detection.
2. For each edge pixel (x_i, y_i), vote for all (a, b) at distance r.
3. Or use gradient: trace along normal to vote for center.

4.3 Gradient-Based Optimization

Use edge gradient to vote only along normal:

$$(a, b) = (x_i, y_i) + r \cdot (\cos\phi, \sin\phi)$$

Reduces to 1D voting per r.

4.4 Example

Circle center $(50, 50)$, $r = 20 \rightarrow$ peak at $A(50, 50, 20)$.

4.5 Advantages

- Robust circle detection.

4.6 Disadvantages

- High memory (3D accumulator).

5 Elliptical Hough Transform

5.1 Concept

Detects ellipses:

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

5 parameters: $(h, k, a, b, \theta) \to$ 5D accumulator (impractical).

5.2 Decomposition Methods

1. Fit ellipse using moments or least squares.
2. Use Hough for major/minor axis, then refine.

5.3 Alternative: Random Sampling

Use RANSAC to fit ellipses from 5 random edge points.

5.4 Advantages

- Detects general conics.

5.5 Disadvantages

- Computationally expensive.

6 Multi-Dimensional Hough Transform

6.1 Concept

Extends Hough to n-parameter shapes:

$$f(x_1, \ldots, x_n; p_1, \ldots, p_n) = 0$$

Accumulator: n-D array.

6.2 Applications

- Motion estimation (3D: v_x, v_y, t).
- Vanishing point detection.

6.3 Curse of Dimensionality

Memory and time grow exponentially with parameters.

7 Randomized Hough Transform (RHT)

7.1 Concept

Samples k points to solve for parameters, votes only if valid.

7.2 Algorithm

1. Randomly pick k edge points.
2. Solve for parameters (e.g., 2 points $\rightarrow$ line).
3. Increment accumulator if valid.
4. Repeat.

7.3 Advantages

- Low memory (no full grid).
- Fast convergence.

8 Distance-Transform-based Hough Transform

8.1 Concept

Use distance transform (DT) to guide voting.

8.2 Method

1. Compute DT of edge map.
2. High DT values $\rightarrow$ likely shape centers.
3. Vote weighted by DT value.

8.3 Applications

- Circle detection (center = high DT).

9 Weighted Hough Transform

9.1 Concept

Assign weights to votes based on edge strength, gradient, or confidence.

$$A(\mathbf{p}) += w_i \quad \text{instead of } +1$$

9.2 Advantages

- Reduces false positives.
- Prioritizes strong edges.

10 3D Hough Transform

10.1 Concept

Detects 3D planes in point clouds:

$$ax + by + cz = d$$

4D accumulator: (a, b, c, d) (normalized).

10.2 Applications

- 3D reconstruction.
- LiDAR data analysis.

11 Hough Transform for Hyperbolas

11.1 Concept

Hyperbola equation:

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

5 parameters $\rightarrow$ 5D voting.

11.2 Use Case

- Pupil detection in eye tracking.

12 Radon Transform (Related to Hough Transform)

12.1 Concept

The **Radon Transform** integrates image intensity along lines at angle θ:

$$R(\rho, \theta) = \int_{-\infty}^{\infty} I(\rho\cos\theta - s\sin\theta, \rho\sin\theta + s\cos\theta)\, ds$$

12.2 Relation to Hough

- Hough: votes in (ρ, θ) for edge points.
- Radon: integrates all points along line.

Inverse Radon $\rightarrow$ reconstruct image (used in CT scanning).

13 Curve Hough Transform

13.1 Concept

Detect arbitrary parametric curves:

$$x = f(t; \mathbf{p}), \quad y = g(t; \mathbf{p})$$

13.2 Example: Parabola

$$y = ax^2 + bx + c$$

3 parameters $\rightarrow$ 3D accumulator.

13.3 Method

Solve for parameters using multiple points.

14 Comparison of Hough Variants

Variant	Parameters	Memory	Speed	Use Case
Standard HT	2	High	Slow	Lines
PHT	2	Medium	Fast	Real-time lines
GHT	Arbitrary	High	Slow	Known shapes
CHT	3	Very High	Slow	Circles
RHT	n	Low	Fast	General
Weighted HT	n	Medium	Medium	Noisy data
3D HT	4	Extreme	Very Slow	3D planes

15 Practical Implementation Tips

- **Quantization:** Choose $\Delta\rho, \Delta\theta$ based on image resolution.
- **Peak Detection:** Use non-maximum suppression in accumulator.
- **Gradient Constraint:** Use edge direction to reduce voting.
- **Multi-Scale:** Detect at multiple resolutions.
- **OpenCV:** `HoughLines`, `HoughLinesP`, `HoughCircles`

16 Conclusion

The Hough Transform family is a cornerstone of classical computer vision. From simple line detection to complex arbitrary shapes, variants of the Hough Transform enable robust parametric feature extraction in noisy and cluttered images. While computationally intensive in high dimensions, modern optimizations (RHT, PHT, gradient-based voting) make it practical for real-time applications.